



Abstract

Cerebral amyloid angiopathy (CAA), which is prevalent among older adults, constitutes a major cause of brain hemorrhage, dementia, and cognitive decline. Currently, no effective treatments for alleviating the disease of CAA exist. Hence, the discovery of novel drug targets for CAA is of utmost significance. Cis-expression quantitative trait locus (cis-eQTL) of druggable genes from eQTLGen Consortium and genome-wide association study of CAA from FinnGen database were used to screen for drug-usable genes causally associated with CAA. Robustness was assessed using colocalization analysis and heterogeneity in dependent instruments (HEIDI) tests. Significant findings were further validated using summary-data-based Mendelian randomization (SMR) with blood and arterial tissue eQTL data from eQTLGen and GTEx. Potential drugs targeting prioritized genes were screened using the DSigDB database, and binding interactions were evaluated via molecular docking. Mendelian randomization analysis identified 10 druggable genes significantly associated with CAA risk (false discovery rate < 0.1). Colocalization analysis revealed strong evidence (PPH4 > 0.75) that dipeptidyl peptidase 9 (DPP9) and CAA share a causal genetic variant. DPP9 is a member of the dipeptidyl peptidase IV family and participates in cellular energy metabolism. SMR with HEIDI test confirmed the causal relationship between genetically predicted DPP9 expression and increased CAA risk in both blood (P_{SMR} < 0.05, P_{HEIDI} > 0.05) and arterial tissues (P_{SMR} < 0.05, P_{HEIDI} > 0.05). Drug screening identified sitagliptin and zoledronic acid as potential DPP9-targeting compounds. Molecular docking demonstrated a highly favorable binding interaction between sitagliptin and DPP9 (binding energy = −9.0 kcal/mol). This study provides genetic evidence identifying DPP9 as a potential drug target for CAA. The antidiabetic drug sitagliptin, predicted to bind DPP9 with high affinity, represents a promising candidate for therapeutic repurposing in CAA.

Abbreviations: Aβ = amyloid-β, CAA = cerebral amyloid angiopathy, cis-eQTL = cis-expression quantitative trait locus, DPP9 = dipeptidyl peptidase 9, GWAS = genome-wide association study, IVW = inverse-variance weighted, MR = Mendelian randomization, SMR = summary-data-based Mendelian randomization, SNP = single-nucleotide polymorphism.

Keywords: cerebral amyloid angiopathy, DPP9, drug targets, Mendelian randomization

1. Introduction

Cerebral amyloid angiopathy (CAA) is a cerebrovascular disorder characterized by the deposition of amyloid-β (Aβ) within the walls of small and medium-sized cortical vessels and leptomeningeal arteries in the brain.^[1] CAA is prevalent among the elderly and represents a significant cause of intracerebral hemorrhage, dementia, and cognitive decline.^[2] As the population ages, CAA is increasing and effective disease-improving interventions are urgently needed. To date, the pathogenesis of CAA remains unclear, despite extensive in vivo and in vitro research, and related treatment methods are still in the exploratory stage.^[3] Thus, the identification of novel drug targets with

the ability to prevent CAA or delay its progression assumes the utmost significance.

Integrating genetics into medication development could be one of the most efficacious strategies for enhancing this process, as therapies supported by genetics have a much higher likelihood of success in clinical trials.^[4] Proteins encoded by druggable genes have emerged as targets for drugs, or potentially for small molecules or monoclonal antibodies.^[5,6] Mendelian randomization (MR) studies can infer the causal effect of exposure on outcomes.^[7]

MR relies on genome-wide association studies (GWAS), which use one or more genetic variants as instrumental variables (IVs)

AX and ZY contributed to this article equally.

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The datasets generated during and/or analyzed during the current study are publicly available.

Since this study was based on previously published articles and open-source databases, ethical approval and informed consent have been previously obtained.

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that are closely associated with the interest exposure and are not affected by confounding factors. In drug target MR analysis, single nucleotide polymorphisms (SNPs) associated with gene expression levels, known as expression quantitative trait loci (eQTL), are utilized as IVs to investigate the effects of druggable genes. Specifically, cis-eQTLs located in genomic regions proximal to the target gene are frequently selected due to their strong association with gene expression.^[8] By integrating cis-eQTLs and CAA risk association data to establish a causal relationship between exposure and outcome, the prediction of drug efficacy is enhanced.

This study employs an integrated analytical pipeline to systematically prioritize and validate novel therapeutic targets for CAA. First, we leverage MR using blood cis-eQTLs as instrumental variables to infer causal relationships between genetically predicted expression of druggable genes and CAA risk. Next, colocalization analysis and summary-data-based MR (SMR) with arterial tissue eQTLs are applied to validate significant signals and distinguish shared causal variants from pleiotropic effects. Finally, we bridge genetic findings to therapeutics by screening the Drug Signatures Database (DSigDB) for candidate compounds targeting prioritized genes, followed by molecular docking to computationally evaluate binding affinity and interactions. This framework synergizes causal genetics, multi-tissue validation, and computational drug discovery to prioritize promising repurposable candidates for CAA therapy.

2. Materials and methods

2.1. Exposure data

We extracted 4463 genes from Finan et al's study,^[6] which focuses on the druggable genome and its support for target identification and validation in drug development, providing a detailed analysis of genes encoding druggable proteins and their potential as drug targets. Blood cis-eQTL data were sourced from the eQTLGen (<https://eqtlgen.org/>), a large-scale research initiative that investigates the genetic architecture of blood gene expression and the genetic basis of complex traits. The eQTLGen data resulted from a genome-wide meta-analysis of whole blood samples from 31,684 healthy individuals of European ancestry, drawn from 37 cohorts, and includes cis-eQTL data for 16,987 genes.^[9] The consortium aims to facilitate collaboration and data harmonization among different blood eQTL cohorts, leverage its high power to detect weak genetic effects on blood gene expression, and provide a valuable resource for interpreting GWAS and developing new methods.

2.2. Outcome data

CAA outcome data were sourced from the nationwide FinnGen Consortium Release R11 (https://storage.googleapis.com/finngen-public-data-r11/summary_stats/finngen_R11_I9_CEREBAMY.gz), which integrates electronic health records with genomic data from Finnish participants collected between 2017 and 2023 under ethical oversight. Cases were defined using the specific ICD-10 diagnostic code I9_CEREBAMY, totaling 101 individuals, while controls consisted of 412,105 population-matched individuals lacking cerebrovascular diagnoses. This dataset provided summary statistics for 16.4 million imputed variants, including association metrics such as odds ratios with 95% confidence intervals and *P*-values, all comprehensively adjusted for covariates including age, sex, and genetic principal components. The exposure and outcome samples were mainly from the European population, thus avoiding population heterogeneity. Furthermore, these samples come from different studies and there is no population overlap, ensuring a valid 2-sample MR Hypothesis.

2.3. IVs selection

The genetic instrumental variables used for MR analysis were subject to 3 MR assumptions: SNPs must be strongly associated with exposure. We selected only cis-eQTLs reaching genome-wide significance ($P < 5.0 \times 10^{-8}$). This stringent threshold ensures strong instruments. SNPs must not be associated with confounders of the exposure–outcome relationship. To ensure selected SNPs were independent (i.e., not in strong linkage disequilibrium [LD]), we performed LD-based clumping using a stringent threshold of $r^2 < 0.1$ within a 10,000kb window. This minimizes the risk of correlated SNPs introducing bias through shared confounding pathways, and SNPs must influence the outcome only through exposure. While this assumption cannot be empirically verified upfront, the selection of cis-eQTLs and subsequent pleiotropy tests help mitigate violations. For each set of SNPs selected for a gene, we calculated the *F*-statistic to assess instrument strength. Only instruments with an *F*-statistic > 10 were retained for MR analysis, indicating a low risk of weak instrument bias.

2.4. MR analysis

To estimate the putative causal effect of blood expression levels of each druggable gene on the risk of CAA. Analyses were performed using the TwoSampleMR package (version 0.5.8) in R. For genes instrumented by a single SNP, the causal effect estimate was calculated using the Wald ratio method (SNP–outcome effect/SNP–exposure effect). For genes instrumented by multiple independent SNPs, the primary analysis used inverse-variance weighted (IVW) regression. IVW provides a consistent causal estimate under the assumption that all selected IVs are valid. Significant heterogeneity among SNP-specific causal estimates can indicate violations of the MR assumptions. Heterogeneity among the causal estimates from individual IVs was assessed using Cochran *Q* test. A *Q* test *P*-value $> .05$ was interpreted as indicating a lack of statistically significant evidence for heterogeneity in the current analysis.^[10] To assess potential directional pleiotropy, we applied MR-Egger regression. A nonsignificant intercept term (*P*-value $> .05$) indicates that our analysis found no statistical evidence for unbalanced horizontal pleiotropy. However, this does not definitively prove the absence of pleiotropy, as the test may lack power to detect small pleiotropic effects, particularly with few genetic instruments.

2.5. Colocalization analysis

For genes showing significant MR results, colocalization analysis aimed to determine if the association between gene expression and CAA was driven by shared causal genetic variants underlying both traits in the same genomic region, rather than distinct variants in LD. We performed colocalization analysis using the coloc^[11] R package. A significant MR result could arise if the eQTL and GWAS signals are correlated but driven by distinct causal variants (LD confounding). Colocalization provides stronger evidence that the same variant influence both traits, bolstering the causal hypothesis from MR. The default prior probability was $P1 = 1E-4$, $P2 = 1E-4$, and $P12 = 1E-5$, respectively representing that a SNP is associated with the expression of the druggable genes, the outcome, or both. The posterior probabilities for the following 5 hypotheses were generated from colocalization analysis: PPH0, no association with either the expression of the druggable genes or CAA; PPH1, association with the expression of the druggable genes but not CAA; PPH2, association with CAA but not the expression of the druggable genes; PPH3, association with the expression of the druggable genes

and CAA, with different causal variants; PPH4, association with the expression of the druggable genes and CAA, with a shared causal variant. A PPH4 > 0.75 was considered strong evidence for colocalization. Genes meeting this threshold were prioritized as high-confidence potential therapeutic targets.

2.6. SMR verification

To independently validate significant gene–CAA associations identified in our primary MR analysis and provide tissue-specific insights, we performed SMR. SMR tests the hypothesis that a genetic variant (SNP) influencing both gene expression and disease risk does so through a shared causal pathway, where altered expression mediates the SNP effect on the disease.^[12] Critically, SMR integrates summary statistics from independent eQTL and GWAS datasets under the MR framework, requiring the core assumption that the SNP affects the outcome only via gene expression. To distinguish true mediation from confounding by distinct causal variants in LD, we applied the Heterogeneity in Dependent Instruments (HEIDI) test. A nonsignificant HEIDI test ($P > .05$) supports a single causal variant driving both signals, strengthening causal inference. We used 2 independent and complementary eQTL reference databases: eQTLGen (whole blood), which aligns with the eQTL source used in our main MR analysis to ensure comparability; and Genotype-Tissue Project (GTEx), focusing on Whole Blood and arterial tissue. GTEx v8 provides genome-wide cis-eQTL associations for 56,000 transcripts and normalized effect sizes for 15,000 protein-coding genes. We selected arterial tissue based on the pathophysiology of CAA, since CAA primarily affects small cerebral arteries, and GTEx lacks data on these, peripheral arteries are the most relevant available tissue for studying vascular biology and may share biological pathways with cerebral vessels. Genes with significant SMR associations ($P_{\text{SMR}} < 0.05$) and no evidence of variant heterogeneity ($P_{\text{HEIDI}} > 0.05$) were considered robustly validated.

2.7. Candidate drug prediction

To identify existing drugs or compounds known to interact with the protein products of genes prioritized through MR and colocalization, thereby nominating them for potential repurposing in CAA. We queried DSigDB (v1.0, <http://dsigdb.tanlab.org/DSigDBv1.0/>), a comprehensive resource that links genes to drugs or compounds based on curated gene sets from various sources. It contains 22,527 gene sets representing interactions with 17,389 distinct compounds, covering 19,531 genes. This database allows systematic identification of compounds known to modulate the activity of our prioritized target genes.

2.8. Molecular docking

To computationally evaluate the binding potential of candidate drugs against protein targets encoded by prioritized genes, we performed molecular docking. Experimentally resolved 3D protein structures were retrieved from the Protein Data Bank (<http://www.rcsb.org/>), representing the highest-quality structural data for human targets or close homologs. Small-molecule drug structures were sourced from the PubChem Compound Database (<https://pubchem.ncbi.nlm.nih.gov/>), ensuring chemical accuracy. Proteins were prepared in PyMOL by removing crystallographic water molecules, ions, and inactive co-crystallized ligands to eliminate steric interference and focus on biologically relevant binding sites. Docking simulations were executed using AutoDock Vina, which predicts ligand binding poses and calculates binding affinity (reported as Gibbs free energy change, ΔG , in kcal/mol). This approach assumes that the lowest-energy conformation represents the most stable protein-ligand complex. The primary outcome, binding energy, serves as a quantitative indicator of interaction strength, where more negative values imply stronger binding. Finally, the top-scoring pose for each drug-target pair was visually analyzed in PyMOL to identify key molecular interactions (e.g., hydrogen bonds, hydrophobic contacts, ionic bonds) within the binding pocket, providing mechanistic insights into druggability.

3. Results

3.1. Mendelian randomization

Based on the IVW approach, we found that the expression of 10 consumable genes was significantly associated with CAA risk at false discovery rate < 0.1 (Fig. 1), and passed the horizontal pleiotropy and heterogeneity tests (Tables 1 and 2, Fig. 2).

3.2. Colocalization

The probability of shared causal variation between cis-eQTL and CAA results were calculated by colocalization analysis. As shown in Table 3, only DPP9 showed strong evidence of colocalization with CAA (PPH4 > 0.75), which may be a candidate drug target gene.

3.3. SMR verification

Notably, DPP9 replicates in 2 gene expression datasets, including eQTLGen (blood), GTEx (blood), and GTEx (arteries). The causal relationship between DPP9 expression and CAA was further verified in blood and arterial tissues. Gene predicted DPP9 expression levels were associated with an increased risk of CAA (Table 4).

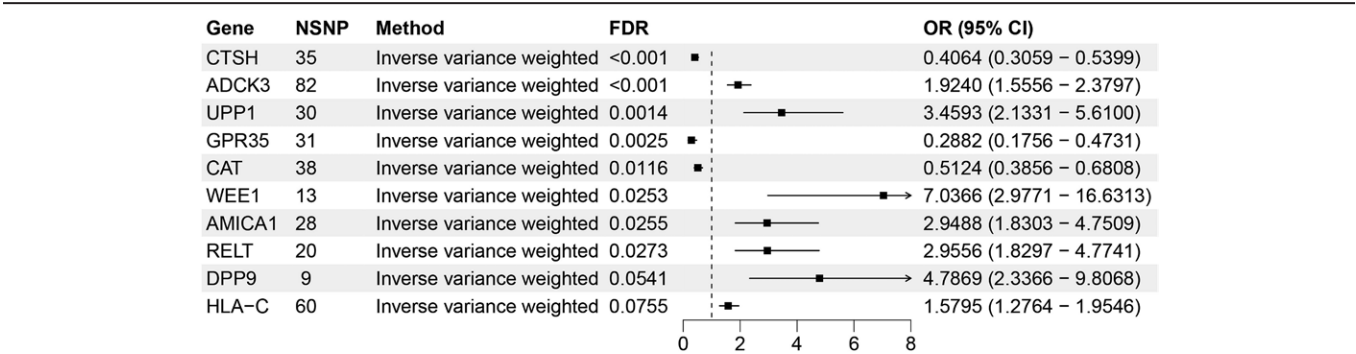


Figure 1. Forest map showing the results of 10 important genes.

Table 1
Heterogeneity results by Cochran Q test of CTSH, ADCK3, UPP1, GPR35, CAT, WEE1, AMICA1, RELT, DPP9, and HLA-C on cerebral amyloid angiopathy.

Exposures	Outcomes	Cochran <i>Q</i> test	Heterogeneity <i>P</i> -value
CTSH	Cerebral amyloid angiopathy	IVW	.725695944
		MR Egger	.688113464
ADCK3		IVW	.955574493
		MR Egger	.955098785
UPP1		IVW	.708872614
		MR Egger	.76041427
GPR35		IVW	.386764653
		MR Egger	.355390401
CAT		IVW	.891463997
		MR Egger	.896586157
WEE1		IVW	.999093041
		MR Egger	.998391037
AMICA1		IVW	.253112683
		MR Egger	.228571114
RELT		IVW	.572902565
		MR Egger	.566547995
DPP9	IVW	.472415846	
	MR Egger	.528732422	
HLA-C	IVW	.507499366	
	MR Egger	.491316001	

IVW = inverse-variance weighted, MR = Mendelian randomization.

3.4. Candidate drug prediction

This study utilized the DSigDB database to make predictions regarding drugs that are likely to be efficacious. As shown in Table 5, only sitagliptin and zoledronic acid were associated with DPP9 based on adjusted *P* values < .05.

3.5. Molecular docking

Molecular docking was used to evaluate the affinity of drug candidates to their targets and to understand the druggability of drug targets. The key components and hub proteins were validated through molecular docking, using a binding energy of −5.0 kcal/mol or lower as the criterion.^[13] A total of 2 effective docking results of proteins with drugs were produced (Table 6 and Fig. 3). DPP9 and sitagliptin showed the lowest binding energy (−9.0 kcal/mol), indicating that the binding was very stable.

4. Discussion

Given that the pathophysiology of CAA remains unknown, the development of novel therapeutic agents for CAA is highly challenging. This study, based on blood and arterial drug eQTL, demonstrated a causal relationship between DPP9 and CAA and showed evidence of colocalization with CAA outcome genes. In addition, we predicted the corresponding drug and carried out molecular docking, which further proved the value of the target gene.

Dipeptidyl peptidase 9 (DPP9), a member of the dipeptidyl peptidase IV family, possesses post-proline dipeptidyl aminopeptidase activity, cleaving Xaa-Pro dipeptides from proteins' N-termini.^[14,15] DPP9 plays an important role in a variety of biological processes and diseases. DPP9 is involved in cellular energy metabolism. Han et al suggested that DPP9 could participate in adipogenesis through the PPARγ2 signaling pathway.^[16] DPP9 is also involved in the regulation of the immune system. The expression of DPP9 is up-regulated in activated lymphocytes, suggesting that DPP9 may be involved in lymphocyte activation.^[17] DPP9 has been

Table 2
Pleiotropy results of MR analysis of CTSH, ADCK3, UPP1, GPR35, CAT, WEE1, AMICA1, RELT, DPP9, and HLA-C on cerebral amyloid angiopathy.

Exposures	Outcomes	SE	Intercept	P-value
CTSH	Cerebral amyloid angiopathy	0.057988007	0.020587283	.724828781
ADCK3		0.048660018	0.043875497	.369936467
UPP1		0.089938884	0.125903637	.172532265
GPR35		0.069535605	−0.041159945	.558486947
CAT		0.053914682	−0.055102193	.313587921
WEE1		0.112740289	0.042627897	.712546016
AMICA1		0.080299672	0.049715337	.541223069
RELT		0.094675866	0.088697546	.361234038
DPP9		0.119715407	0.147387229	.258023802
HLA-C		0.086056199	−0.064636484	.455631523

shown to be expressed in many different types of tumors, such as brain tumors, gynecological malignancies, liver cancer, and hematological malignancies.^[18] In vascular diseases, DPP9 has been reported to be abundant in macrophage-rich areas of atherosclerotic plaques. DPP9 promotes the activation of pro-inflammatory M1 macrophages, which play a key role in atherogenesis, plaque rupture, and subsequent infarction.^[19] The expression of the dipeptidyl peptidase IV family in endothelial cells is highly regulated and has spatial heterogeneity in the cardiovascular tree. For example, the expression of DPP8 in cardiac microvascular endothelium is significantly higher than that of other endothelial cells, while DPP9 is the major DPP in human carotid endothelial cells.^[20] Dipeptidyl-peptidase 4 (DPP-4) activity is known to increase early in the brain in sporadic Alzheimer disease (AD) and is associated with Aβ plaques.^[21] We speculate that DPP9 may be expressed in cerebrovascular endothelial cells and participate in the deposition of Aβ in the cerebrovascular system.

CAA affects approximately 25% of the elderly and coexists with AD in at least 50% of cases.^[22] Microvascular degeneration caused by CAA leads to hypoperfusion, neuroinflammation, blood–brain barrier dysfunction, and oxidative stress, further promoting the neurodegenerative process^[23] and posing a significant risk of vascular lesions in AD and other dementias. The perivascular accumulation of Aβ in CAA leads to cerebral endothelial cell dysfunction and death.^[23] Sitagliptin is a DPP-4 inhibitor, and DPP9 was also found to be its therapeutic target through molecular docking. Shalmon et al found that for type 2 diabetes patients with both Alzheimer disease and other dementias, sitagliptin is superior to linagliptin and vildagliptin when using DPP-4 inhibitors, because linagliptin and vildagliptin have a strong correlation with drug-related blood pressure risks, while there is no association between sitagliptin administration and blood pressure.^[24] Sitagliptin was recently reported to reduce Aβ accumulation in a mouse model of type 2 diabetes combined with Alzheimer disease by increasing levels of the insulin-degrading enzyme.^[25] The use of antidiabetic drugs to treat AD has become a promising treatment strategy. For example, GLP-1 agonists are currently undergoing human trials in Alzheimer disease, with preclinical findings suggesting potential benefits in cognitive function and disease progression. Given that CAA shares pathophysiological features with AD, the antidiabetic drug sitagliptin represents a promising avenue for developing new treatments for CAA.

We recognize the limitations of the study. Firstly, the study was confined to a Finnish population, resulting in race-specific conclusions that might not have been verified in other

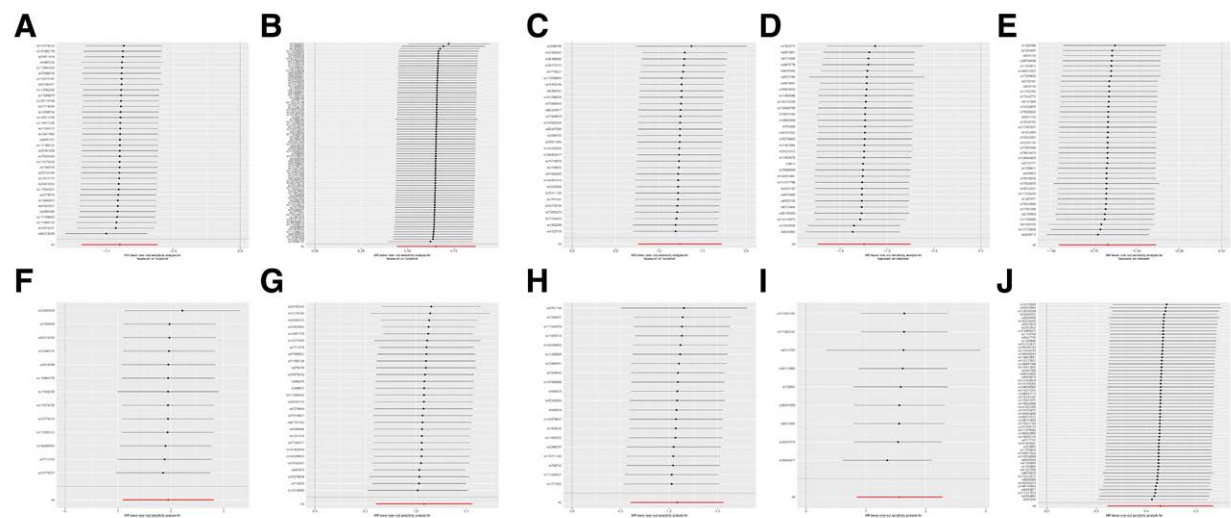


Figure 2. Leave-one-out analysis visualizing the Mendelian randomization (MR) estimates of the different exposures (A: CTSH, B: ADCK3, C: UPP1, D: GPR35, E: CAT, F: WEE1, G: AMICA1, H: RELT, I: DPP9, J: HLA-C) with the outcome(cerebral amyloid angiopathy).

Table 3
Colocalization results of 10 genes.

Gene	PPH0	PPH1	PPH2	PPH3	PPH4
CTSH	0	0.636441058	0	0.002472089	0.361086853
ADCK3	0	0.92058389	0	0.008036447	0.071379663
UPP1	0	0.522378278	0	0.001676267	0.475945454
GPR35	1.42E–251	0.422903928	8.20E–254	0.001859796	0.575236276
CAT	0	0.903900384	0	0.003668903	0.092430713
WEE1	1.61E–120	0.801779396	2.53E–123	0.001061479	0.197159126
AMICA1	0	0.731170464	0	0.002140531	0.266689005
RELT	3.66E–321	0.553894602	9.88E–324	0.001131957	0.44497344
DPP9	3.05E–141	0.227735684	1.68E–143	0.000484116	0.771780199
HLA-C	0	0.848988378	0	0.005001992	0.14600963

PPH0-PPH4 represents the posterior probabilities conform to 1 of the 5 hypotheses.

Table 4
SMR results.

Gene	Tissue	b_SMR	se_SMR	p_SMR	p_HEIDI	nsnp_HEIDI
DPP9	GTEx(Blood)	1.66707	0.512763	1.15E–03	5.16E–01	15
DPP9	GTEx(Artery)	1.89453	0.63744	2.96E–03	6.67E–01	9
DPP9	eQTLGen(Blood)	1.55736	0.459789	7.06E–04	2.94E–01	20

Table 5
Candidate drug predicted using DSigDB.

Drug names	P value	Adjusted P value	Genes
Sitagliptin	.001221561	.002443121	DPP9
Zoledronic acid	.02351504	.02351504	DPP9

Table 6
Docking results showed available proteins with small molecules.

Target	PDB ID	Drug	PubChem ID	Binding energy
DPP9	7ZXS	Sitagliptin	4369359	–9
DPP9	7ZXS	Zoledronic acid	68740	–5.6

ethnic groups. Furthermore, the paucity of diversity in study cohorts can lead to the omission of significant genetic variants related to other populations, potentially influencing the efficacy of identified drug targets among different ethnicities. Secondly, the derivation of core genes is maintained at the genomic level. To fully illuminate the specific function of DPP9 in CAA, further exploration of transcription and translation mechanisms is requisite. It is necessary for other researchers to validate our conclusions through multi-omics studies.

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